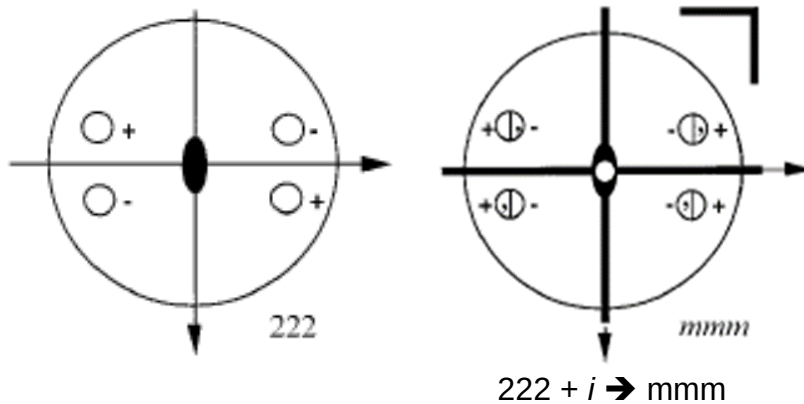
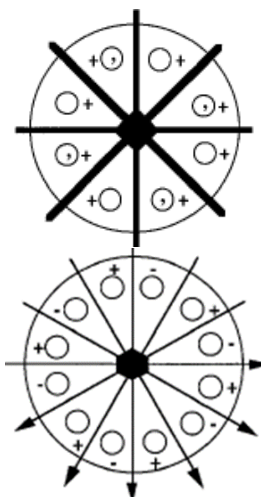


Q1) 3D Point Groups: (20 pts)

a) Using a projection sketch, determine which point group results from adding a center of symmetry to 222



b) Use the following projections to identify the point group and crystal system, describe the symmetry elements (type and orientation) and order of the group.



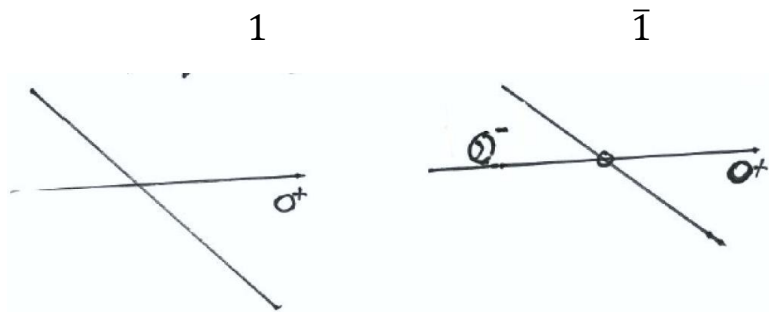
Point group	Crystal system	Symmetries	Group order
4mm	Tetragonal	Tetrad along c-axis Mirror plane normal to x and y axes Mirror plane normal to axis bisecting x-y plane	8
622	Hexagonal	Hexad along c-axis Diads 120° apart in basal plane Diads bisecting axes in basal plane	12

Q2) Point and space groups: (40 pts)

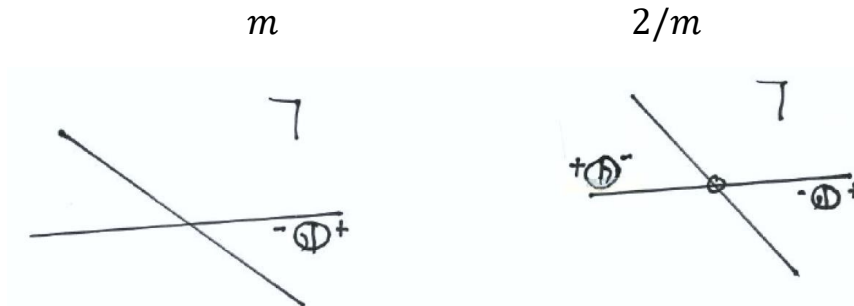
- a) Using sketches, determine which point groups result from adding a center of symmetry to the following point groups: 1 , m , $mm2$, $4mm$, 6 , $\bar{6}$, $6mm$.

Answer: In each case, note that adding a center of symmetry (inversion) to the symmetry elements takes all coordinates from x, y, z to $-x, -y, -z$, and changes the handedness.

$$1 + i = \bar{1}$$



$$m + i = 2/m$$

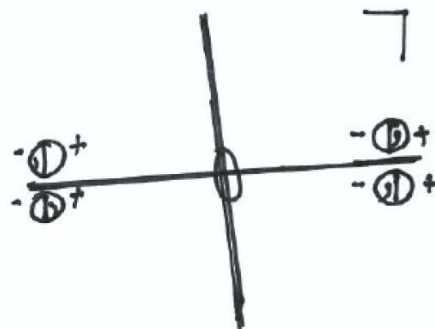
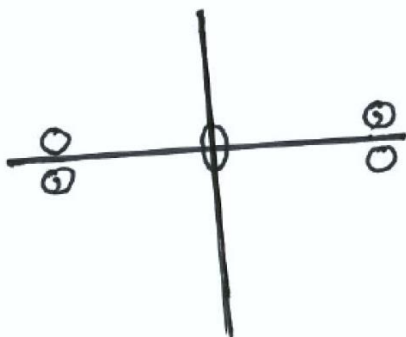


$$mm2 + i = mmm$$

$mm2$

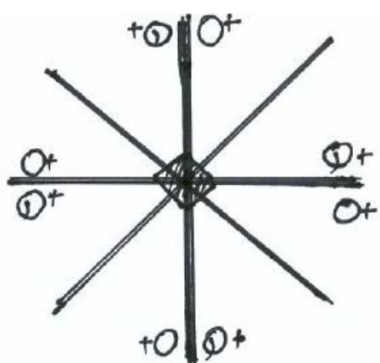
mmm

$$mm2 + i = mmm$$

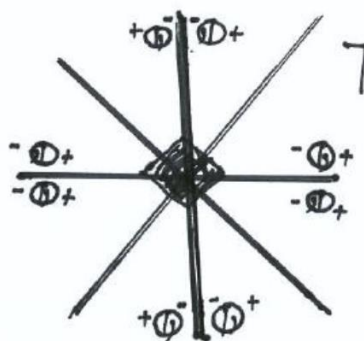


$$4mm + i = 4/mmm$$

4mm

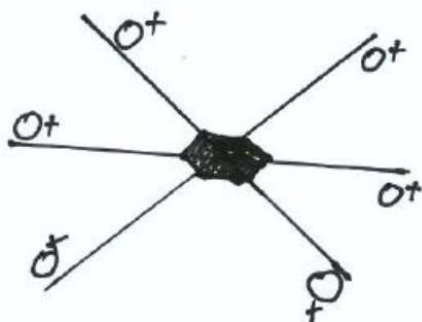


4/mmm

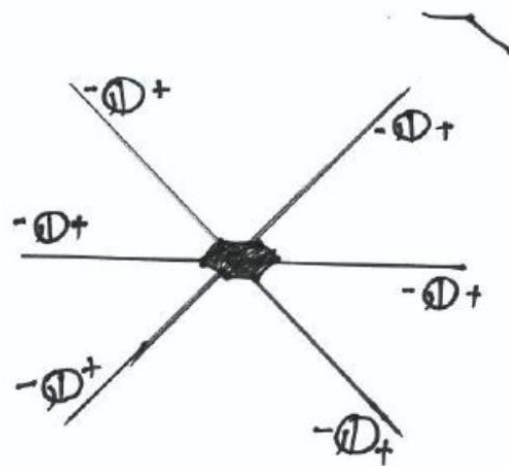


$$6 + i = 6/m$$

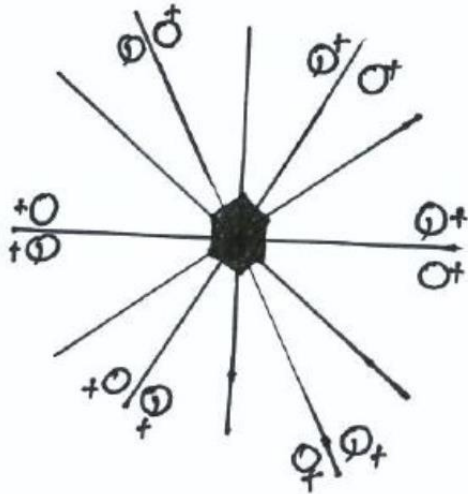
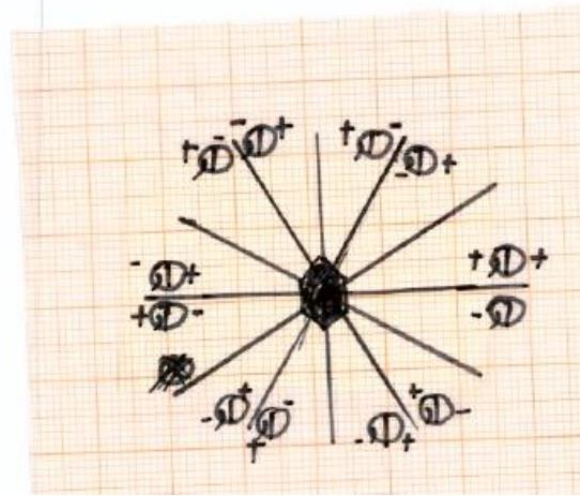
6



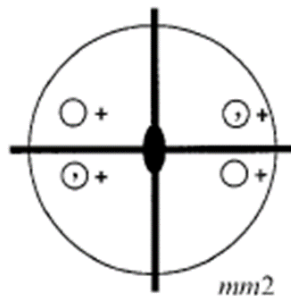
6/m



$$6mm + i = 6/mmm$$

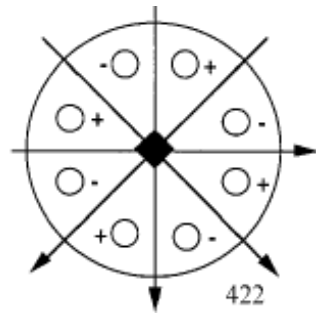
 $6mm$

 $6/mmm$


- b) None of the original point groups are centrosymmetric (absence of $\bar{1}$ and $3\bar{2}$ and $4/m$ or $6/m$). All point groups are centrosymmetric after addition of center of symmetry.
- c) Use a diagram to determine what point group results from combination of:
- Two mirror planes oriented at 90° with respect to one another and a diad parallel to both.



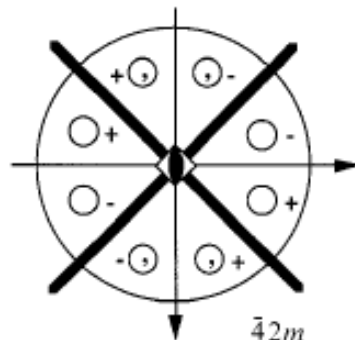
Answer:

b. A tetrad axis with a perpendicular diad.



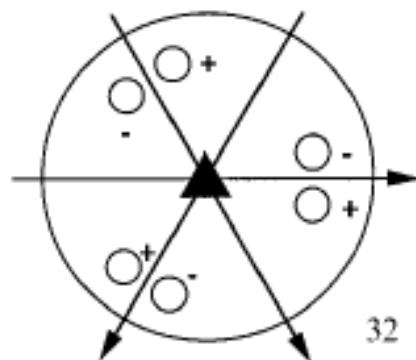
Answer:

c. A 4-bar axis with a mirror oriented with its normal perpendicular to the roto-inversion axis



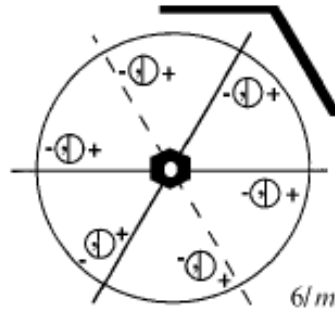
Answer:

d. A triad axis with a perpendicular diad



Answer:

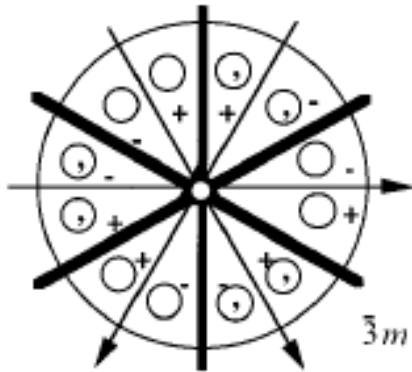
e. A hexad axis with a perpendicular mirror



Answer: .

- d) Does the point group $\bar{3}m$ contain symmetry elements not present in the name?

Answer: Yes, it contains diads at 60° .



- e) Specify the point group and Bravais lattices of the crystals that have the following space groups: $P\bar{4}2_1c$, $Ccc2$, $P6_3mc$

Answer:

$P\bar{4}2_1c \rightarrow \bar{4}2m$ tetragonal primitive

$Ccc2 \rightarrow mm2$ Orthorhombic c

$P6_3mc \rightarrow 6mm$ hexagonal primitive

- f) $\bar{4}2m$, $mm2$ and $6mm$ all lack odd rotoinversions and even rotations with perpendicular mirror plane, therefore they are all centrosymmetric.

Q3) General coordinates of point groups and space groups (20 pts)

- a) Using the matrix approach, determine the general coordinates of the point group 4 and the order of the point group.

Tetrad rotations: 90° , 180° and 270° :

Refer to in-class exercise #4 with $z \neq 0$ for calculations.

$$(x, y, z) \rightarrow (y, -x, z), (-x, -y, z), (-y, x, z)$$

- b) Determine the general coordinates of the point group $4mm$ based on your answer in a) and the order of the point group.

$4mm$ is a tetragonal point group consisting of a tetrad along z , mirror planes normal to x and y axes and mirror planes normal to the axes bisecting the x and y axes.

We can use a mirror plane normal to the x axis and going through the center of the point group to reflect all 4 points, which changes the sign in front of the first coordinate:

$$1 (x, y, z) \rightarrow 5 (-x, y, z)$$

$$2 (y, -x, z) \rightarrow 6 (-y, -x, z)$$

$$3 (-x, -y, z) \rightarrow (-(-x), -y, z) = 7 (x, -y, z)$$

$$4 (-y, x, z) \rightarrow (-(-y), x, z) = 8 (y, x, z)$$

We now have 8 points and no duplicates.

Next, we can use the other mirror plane (normal to the y axis) with a few of these coordinates to see whether additional points are generated or duplicates are formed:

Test:

$$1 (x, y, z) \rightarrow 7 (x, -y, z) \quad \text{Duplicate? Y}$$

$$5 (-x, y, z) \rightarrow 3 (-x, -y, z) \quad \text{Duplicate? Y}$$

$$2 (y, -x, z) \rightarrow 8 (y, x, z) \quad \text{Duplicate? Y}$$

$$6 (-y, -x, z) \rightarrow 4 (-y, x, z) \quad \text{Duplicate? Y}$$

Therefore the order of the group is 8.

- c) What are the general coordinates of $I4mm$? What is the order of the space group?

$I4mm$ is the space group corresponding to tetragonal I , which implies a centering operation $(+\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2})$

The general coordinates of $I4mm$ are:

$(x, y, z), (-x, y, z), (y, -x, z), (-y, -x, z), (-x, -y, z), (x, -y, z), (-y, x, z), (y, x, z),$
 $(x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}), (-x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}), (y+\frac{1}{2}, -x+\frac{1}{2}, z+\frac{1}{2}), (-y+\frac{1}{2}, -x+\frac{1}{2},$
 $z+\frac{1}{2}), (-x+\frac{1}{2}, -y+\frac{1}{2}, z+\frac{1}{2}), (x+\frac{1}{2}, -y+\frac{1}{2}, z+\frac{1}{2}), (-y+\frac{1}{2}, x+\frac{1}{2}, z+\frac{1}{2}), (y+\frac{1}{2},$
 $x+\frac{1}{2}, z+\frac{1}{2}).$

The order of the space group is 16.

Q4) Bravais lattices and space groups: (30 pts)

One common structure for compounds with the ABO_4 stoichiometry is Scheelite, typified by $CaWO_4$.

Table 3B.3. *The structure of scheelite, $CaWO_4$.*

Formula unit:	$CaWO_4$
Space group:	$I4_1/a$ (no. 88)
Cell dimensions:	$a = 5.24 \text{ \AA}, c = 11.38 \text{ \AA}$
Cell contents:	4 formula units
Atomic positions:	
Ca (4b)	$(0,0,1/2); (1/2,0,1/4); +I$
W (4a)	$(0,0,0); (0,1/2,1/4); +I$
O (16f)	$(x, y, z); (\bar{x}, \bar{y}, z); (x, y + 1/2, 1/4 - z); (\bar{x}, 1/2 - y, 1/4 - z);$ $(\bar{y}, x, \bar{z}); (y, \bar{x}, \bar{z}); (\bar{y}, x + 1/2, z + 1/4); (y, 1/2 - x, z + 1/4); +I$ $x = 0.25, y = 0.15, z = 0.075$

- a) What is the Bravais lattice of this crystal?

Answer: tetragonal I

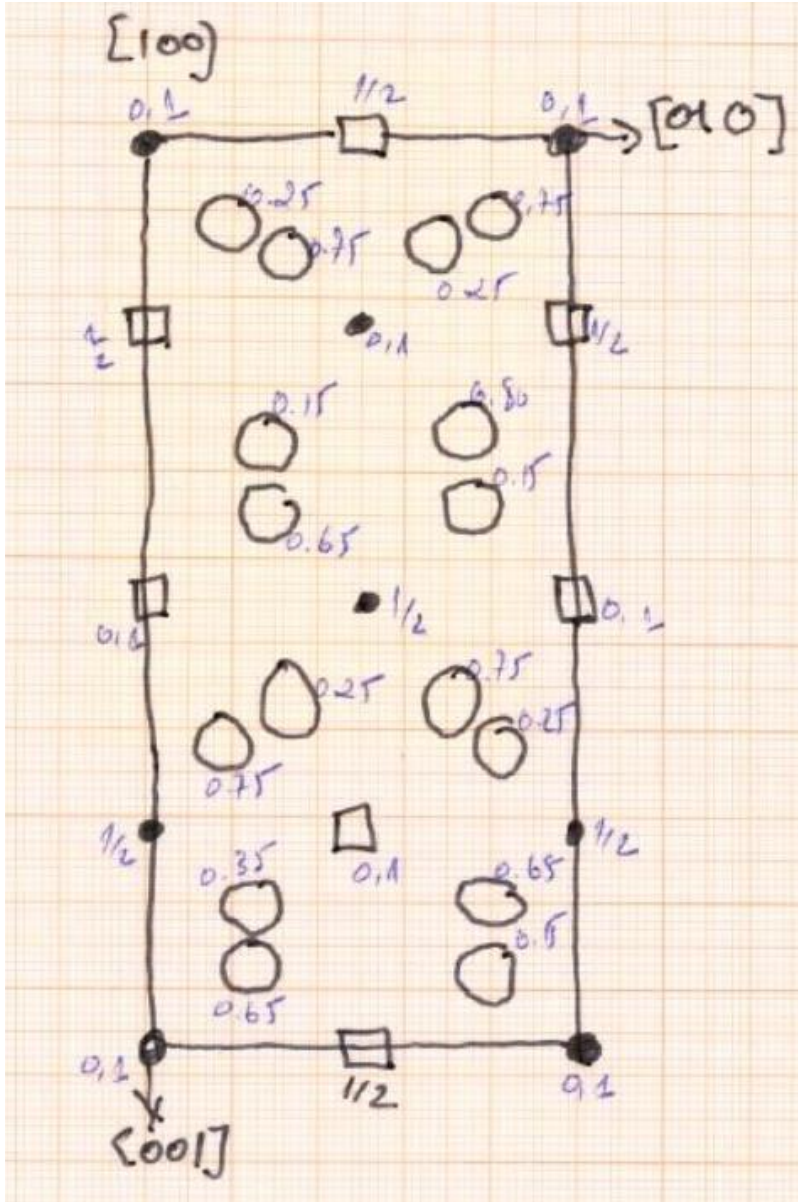
- b) Is scheelite centrosymmetric or non-centrosymmetric?

Answer: the point group is $4/m$, therefore it is centrosymmetric

c) Sketch a projection down $[100]$

Solution:

Legend: Ca: open square; O: large open circle; W: small solid circle



d) Describe the coordination of the W atoms and calculate the bond distances

Solution: W is coordinated with 4 oxygen atoms.

Bond distance: We will calculate one W-O distance. Let's take the W atom at origin $(0,0,0)$ and the O atom at (x,y,z) , where $x=0.25$, $y=0.15$ and $z=0.075$.

Therefore, we can assume $\Delta x=0.25$, $\Delta y=0.15$, and $\Delta z=0.075$. We also note

that $a = b = 5.24 \text{ \AA}$, $c = 11.38 \text{ \AA}$. Since all angles in a tetragonal crystal are normal, we can use Pythagoras to calculate the distance in the given units.

$$L = \sqrt{(0.25 \times 5.24 \text{ \AA})^2 + (0.15 \times 5.24 \text{ \AA})^2 + (0.075 \times 11.38 \text{ \AA})^2} = \sqrt{3.063 \text{ \AA}^2} = 1.75 \text{ \AA}$$

e) Are the general positions of this group occupied?

Solution: This requires knowledge about the order of the space group and the multiplicity of the most general coordinate.

In this case, O is in 16f, which includes the coordinate (x,y,z) , which is the most general coordinate. $I4_1/a$ takes (x,y,z) and applies a screw tetrad, which creates 4 distinct general coordinates. Assuming these do not overlap with the glide plane, the latter produces 4 additional points. The centering then takes all 8 and adds $1/2, 1/2, 1/2$ to create 16 general coordinates. This is the order of the group and cannot be surpassed, hence the answer is *yes, the general coordinates of the space group are occupied by O*.

f) What is the point symmetry of the site occupied by W?

Answer: Normally, general coordinates do not sit on point symmetries. However, W occupies the 4a sites. The Wyckoff letter is the lowest in the alphabet, therefore W occupies a site that corresponds with the highest symmetry of the space group. In $I4_1/a$, the screw tetrad is the highest symmetry. Therefore all *W atoms are on a screw tetrad*.

Alternatively, one can answer that in a tetragonal system the tetrad (and screw tetrad) is always along the c-axis and centered on the unit cell's a-b plane. Given the coordinates of W include the body center, namely, $(1/2, 1/2, 1/2)$, this means that sites occupied by W have the screw tetrad.

g) With respect to the atoms in the unit cell, where is the glide plane?

Answer: $I4_1/a$ implies a glide plane normal to the screw tetrad. The glide plane is therefore parallel to the $[001]$ plane. The translations are obviously in the glide plane, axial and perform a translation of $1/2$ along $[100]$ or $[010]$ directions ($a=b$). The glide planes lie at $z = 3/8, 1/2, 5/8, 7/8$.